

Math 151A Homework 4

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Q1. Interpolation at $x = 0.45$ with $x_0 = 0$, $x_1 = 0.6$, $x_2 = 0.9$

3.1 Q1.

For each f , let P_1 be the linear interpolant through $(x_0, f(x_0))$ and $(x_1, f(x_1))$. Let P_2 be the quadratic Lagrange interpolant through the three nodes.

Denominators used once:

$$(x_0 - x_1)(x_0 - x_2) = 0.54, \quad (x_1 - x_0)(x_1 - x_2) = -0.18, \quad (x_2 - x_0)(x_2 - x_1) = 0.27.$$

At $x = 0.45$,

$$(x - x_1)(x - x_2) = 0.0675, \quad (x - x_0)(x - x_2) = -0.2025, \quad (x - x_0)(x - x_1) = -0.0675.$$

Hence

$$P_2(0.45) = f(x_0) \frac{0.0675}{0.54} + f(x_1) \frac{-0.2025}{-0.18} + f(x_2) \frac{-0.0675}{0.27} = 0.125 f(x_0) + 1.125 f(x_1) - 0.25 f(x_2).$$

The linear value is

$$P_1(0.45) = f(x_0) \frac{0.45 - 0.6}{0 - 0.6} + f(x_1) \frac{0.45 - 0}{0.6 - 0} = 0.25 f(x_0) + 0.75 f(x_1).$$

(a) $f(x) = \cos x$.

$$f(0) = 1, \quad f(0.6) \approx 0.8253356, \quad f(0.9) \approx 0.6216099, \quad f(0.45) \approx 0.9004471.$$

$$P_1(0.45) \approx 0.869002, \quad |P_1 - f| \approx 3.14 \times 10^{-2}.$$

$$P_2(0.45) \approx 0.898100, \quad |P_2 - f| \approx 2.35 \times 10^{-3}.$$

(b) $f(x) = \sqrt{1+x}$.

$$f(0) = 1, \quad f(0.6) \approx 1.2649111, \quad f(0.9) \approx 1.3784049, \quad f(0.45) \approx 1.2041590.$$

$$P_1(0.45) \approx 1.198683, \quad |P_1 - f| \approx 5.48 \times 10^{-3}.$$

$$P_2(0.45) \approx 1.203424, \quad |P_2 - f| \approx 7.35 \times 10^{-4}.$$

(c) $f(x) = \ln(1+x)$.

$$f(0) = 0, \quad f(0.6) \approx 0.4700036, \quad f(0.9) \approx 0.6418539, \quad f(0.45) \approx 0.3711007.$$

$$P_1(0.45) \approx 0.352503, \quad |P_1 - f| \approx 1.86 \times 10^{-2}.$$

$$P_2(0.45) \approx 0.368291, \quad |P_2 - f| \approx 2.81 \times 10^{-3}.$$

(d) $f(x) = \tan x$.

$$f(0) = 0, \quad f(0.6) \approx 0.684136, \quad f(0.9) \approx 1.260158, \quad f(0.45) \approx 0.483000.$$

$$P_1(0.45) \approx 0.513102, \quad |P_1 - f| \approx 3.01 \times 10^{-2}.$$

$$P_2(0.45) \approx 0.454614, \quad |P_2 - f| \approx 2.84 \times 10^{-2}.$$

Code used to generate the numbers:

```
import math
x0, x1, x2, x = 0.0, 0.6, 0.9, 0.45

def P1(f):
    return f(x0)*(x-x1)/(x0-x1) + f(x1)*(x-x0)/(x1-x0)

def P2(f):
    L0 = (x-x1)*(x-x2)/((x0-x1)*(x0-x2))
    L1 = (x-x0)*(x-x2)/((x1-x0)*(x1-x2))
    L2 = (x-x0)*(x-x1)/((x2-x0)*(x2-x1))
    return f(x0)*L0 + f(x1)*L1 + f(x2)*L2

funcs = {
    "cos": math.cos,
    "sqrt1px": lambda t: math.sqrt(1+t),
    "ln1px": lambda t: math.log(1+t),
    "tan": math.tan
}

for name,f in funcs.items():
    true = f(x)
    lval = P1(f)
    qval = P2(f)
    print(name, true, lval, abs(lval-true), qval, abs(qval-true))
```

Q2. Barycentric form

Let the Lagrange basis be

$$\ell_j(x) = \frac{\Psi_n(x)}{(x - x_j)\Psi'_n(x_j)}, \quad \Psi_n(x) = \prod_{k=0}^n (x - x_k).$$

Then the interpolant is

$$p_n(x) = \sum_{j=0}^n f(x_j) \ell_j(x) = \Psi_n(x) \sum_{j=0}^n \frac{f(x_j)}{(x - x_j)\Psi'_n(x_j)}.$$

Also

$$\sum_{j=0}^n \ell_j(x) = \Psi_n(x) \sum_{j=0}^n \frac{1}{(x - x_j) \Psi'_n(x_j)} = 1.$$

Define weights $w_j = (\Psi'_n(x_j))^{-1}$. Divide the first identity by the second to eliminate $\Psi_n(x)$:

$$p_n(x) = \frac{\sum_{j=0}^n \frac{w_j f(x_j)}{x - x_j}}{\sum_{j=0}^n \frac{w_j}{x - x_j}}, \quad x \neq x_j,$$

which is the barycentric representation. :contentReference[oaicite:0]index=0

Q3. Vandermonde interpolation and evaluation

Data: $(0, 1)$, $(-1, 2)$, $(1, 3)$. Seek $p(x) = a_0 + a_1x + a_2x^2$.

From the data:

$$\begin{cases} a_0 = 1, \\ a_0 - a_1 + a_2 = 2, \\ a_0 + a_1 + a_2 = 3. \end{cases} \Rightarrow a_2 = \frac{3}{2}, a_1 = \frac{1}{2}, a_0 = 1.$$

So

$$p(x) = \frac{3}{2}x^2 + \frac{1}{2}x + 1, \quad p(0.73) = \frac{3}{2}(0.73)^2 + \frac{1}{2}(0.73) + 1 \approx 2.16435.$$

Code that builds the Vandermonde system and evaluates $p(0.73)$:

```
import numpy as np

# data
xs = np.array([0.0, -1.0, 1.0])
ys = np.array([1.0, 2.0, 3.0])

# Vandermonde for degree 2 in monomial basis [1, x, x^2]
V = np.vstack([np.ones_like(xs), xs, xs**2]).T
a0, a1, a2 = np.linalg.solve(V, ys) # coefficients

def p(t):
    return a0 + a1*t + a2*t*t

print("coeffs:", a0, a1, a2)
print("p(0.73) =", p(0.73))
```

4.

I looked at a practical way to evaluate the Lagrange interpolant using the barycentric formula. It writes the interpolant as

$$p_n(x) = \frac{\sum_{j=0}^n \frac{w_j f(x_j)}{x - x_j}}{\sum_{j=0}^n \frac{w_j}{x - x_j}}, \quad w_j = \left(\prod_{k \neq j} (x_j - x_k) \right)^{-1}.$$

This is just the same polynomial from Section 3.1 written in a form that is easy to compute and numerically stable. I also noted that choosing Chebyshev nodes instead of evenly spaced nodes usually reduces the oscillations near the ends of the interval, which matches the accuracy concerns in Section 3.2.

I chose this topic because it is a small step beyond what we do in class, it is simple to implement, and it directly improves accuracy and speed without learning a new method.